

CHAPTER 2

IRT Models for the Analysis of Polytomously Scored Data

Brief and Selected History of Model Building Advances

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Introduction

Editor Introduction: This chapter places the models developed in later chapters into a common historical context. The authors review dichotomous IRT models and lay an important foundation for the concept of information—a key concept in IRT. They also discuss nonparametric IRT and provide an introduction to the issue of parameter estimation. This provides an excellent starting point from which to delve deeper into the specific models, issues surrounding these models, and uses of the models that are provided in later chapters.

The publication of several papers by Lord (1952, 1953a, 1953b) marked the beginning of the transition from classical to modern test theory and practices—a response, in part, to Gulliksen's (1950) challenge to develop invariant item statistics for test development. Modern test theory is characterized by strong modeling of the data, and modeling examinee responses or scores at the item level. Item response theory, originally called latent trait

theory and later item characteristic curve theory, is about linking examinee item responses or scores via item response functions to the latent trait or traits measured by the test. According to Baker (1965), Tucker (1946) may have been the first to use the term *item characteristic curve*, and Lazarsfeld (1950) appears to have been the first to use the term *latent trait* (Lord, personal communication, 1977). In his 1980 monograph, Lord coined the terms *item response function* (IRF) and *ability* as alternatives for *item characteristic curve* and *trait*.

Progress in model building and model parameter estimation was slow initially—almost certainly because of the mathematical complexity of the modeling and the computational demands required for model parameter estimation. There was also considerable skepticism among some researchers about any measurement advantages that might accrue from the modeling. This skepticism remained well into the 1970s. The skeptics were particularly concerned about the strong model assumptions that needed to be made (e.g., unidimensionality), and secondarily the computational challenges that item response theory (IRT) modeling posed. But the promise of these IRT models was great (e.g., model parameter invariance; possibility to deal with missing data designs, for instance, in equating studies or adaptive testing) and the research continued at an exponential rate in the 1970s and 1980s.

The real turning point in the transition process came with the publication of *Statistical Theories of Mental Test Scores* by Lord and Novick (1968). The transition was helped along considerably by Rasch (1960) and work in the late 1950s by Birnbaum (1957, 1958).

Today, there is widespread use of IRT models with both binary and polytomously scored data, although the latter is less well developed and understood. Hence the need for this book and others like it (e.g., van der Linden & Hambleton, 1997). Until the mid-1980s, test developers, psychologists, and researchers with polytomous response data tended to do classical scaling—as outlined by Thurstone (1925). This meant assuming a normal distribution—certainly unrealistic in many applications. Some researchers simply dichotomized their data, even though one of the consequences may be loss of fit of the model (Jansen & Roskam, 1983; Roskam & Jansen, 1986). Lack of software limited IRT applications using polytomous response data, though Andrich (1978) and later Thissen (1981) moved things forward with software for selected polytomous response IRT models.

Interestingly, although Lord and Novick (1968) provided a formulation of a general theory of multidimensional ability spaces, there had been no development up to that time with specific models for the analysis of polytomous response data. Perhaps this was because both authors were working in the educational testing area, where in 1968, binary data were much more common.

The purposes of this chapter will be (1) to introduce many of the polytomous response IRT models that are available today, including several that are the focus of this book, (2) to provide background for the motivations of model

developers, and (3) to highlight similarities and differences among the models, and challenges that still remain to be addressed for successful applications.

It is interesting to note that polytomous response IRT models were introduced long before they found any use in education and psychology, albeit without the necessary software to implement the models. We will begin by introducing Samejima's (1969, 1972) work in the late 1960s with the graded response model, the free response model, and multidimensional models too. Her work was followed by Bock (1972) and the nominal response model—here, the multiple score categories did not even need to be ordered. Later, advances came from Andrich (1978, 1988) with the rating scale model, and this model did receive some fairly quick use, in part because he made software available. Other models to follow Samejima's pioneering work included the partial credit model and variations (e.g., Andersen, 1973; Masters & Wright, 1984; Tutz, 1997; Verhelst, Glas, & de Vries, 1997), the generalized partial credit model (e.g., Muraki, 1992), as well as models by Embretson, Fischer, McDonald, and others for applying multidimensional IRT models to polytomous response data. By 1997, when van der Linden and Hambleton published their edited book *Handbook of Modern Item Response Theory* (1997), they reported the existence of over 100 IRT models and organized them into six categories: (1) models for items with polytomous response formats, (2) nonparametric models, (3) models for response time or multiple attempts on items, (4) models for multiple abilities or cognitive components, (5) models for nonmonotone items, and (6) models with special assumptions about the response models. Only models in the first two categories will be described in this chapter. Readers are referred to van der Linden and Hambleton (1997) for details on models in all six categories.

Development of IRT Models to Fit Polytomously Scored Data

Some of the seminal contributions to the topic of IRT model development are highlighted in Figure 2.1.

We will begin our selected history of IRT model development by focusing first on those models that were developed to handle binary-scored data such as multiple-choice items and short-answer items with achievement and aptitude tests, and true-false, yes-no type items with personality tests. These developments laid the foundation for those that followed for polytomous response data, as the models are based on the same assumptions of unidimensionality and statistical independence of item responses, and model parameters are similar in their purpose and interpretation. Parameter estimation methods were simply extended to handle the extra model parameters with the polytomous response data.

IRT Models to Fit Binary-Scored Data

A primary reason for IRT's attractive features is that explicit, falsifiable models are used in developing a scale on which test items and examinees

Year	Model and the Developer
1952	one- and two-parameter normal-ogive models (Lord)
1957–1958	two- and three-parameter logistic models (Birnbaum)
1960	one-parameter logistic model (Rasch)
1961	Rasch rating scale model (Rasch)
1967	normal-ogive multidimensional model (McDonald)
1969	two-parameter normal ogive and logistic graded response model (Samejima)
1969	multidimensional model (Samejima)
1972	continuous (free) response model (Samejima)
1972	nominal response model (Bock)
1973	Rasch rating scale model (Andersen)
1976	linear logistic Rasch model (Fischer)
1978	Rasch rating scale model (Andrich's full development was carried out and independently of the work of Andersen)
1980	multi-component response models (Embretson)
1981	four-parameter logistic model (Barton & Lord)
1982	partial credit model (Masters)
1985	linear logistic multidimensional model (Reckase)
1988	unfolding model (Andrich)
1990	sequential step model (Tutz)
1991	non-parametric models (Ramsey)
1992	generalized partial credit model (Muraki)
1992	full information item bifactor model (Gibbons & Hedeker)
1993	steps model (Verhelst & Glas)

FIGURE 2.1 The most popular of the unidimensional and multidimensional models for analyzing binary-scored and polytomously scored data.

are placed (Baker, 1965; Hambleton, Swaminathan, & Rogers, 1991; van der Linden & Hambleton, 1997). All IRT models define the probability of a positive response as a mathematical function of item properties, such as difficulty, and examinee properties, such as ability level. For example, one of the popular models used with dichotomous item response data is the three-parameter logistic model (3PLM), expressed as

$$P(u_{ij} = 1|\theta_j) = c_i + (1 - c_i) \frac{\exp[a_i(\theta_j - b_i)]}{1 + \exp[a_i(\theta_j - b_i)]} \tag{2.1}$$

Here $P(u_{ij} = 1|\theta_j)$ indicates the probability of a correct response for examinee j . Hereafter, $P(u_{ij} = 1|\theta_j)$ will be written as $P(\theta)$. The ability parameter for person j is denoted θ_j and, although theoretically unbounded, ranges from -3.0 to 3.0 for a typical population with ability estimates scaled to a mean of zero and a standard deviation of 1.0 , where larger positive values indicate higher ability. The lower asymptote is denoted c_i , also known as the guessing parameter; in other words, the c parameter indicates the probability of positively endorsing an item for examinees with very low ability levels. The difficulty parameter for item i is denoted b_i . The b parameter is on the same scale as θ and is defined as the θ -value where $P(\theta)$ is halfway between

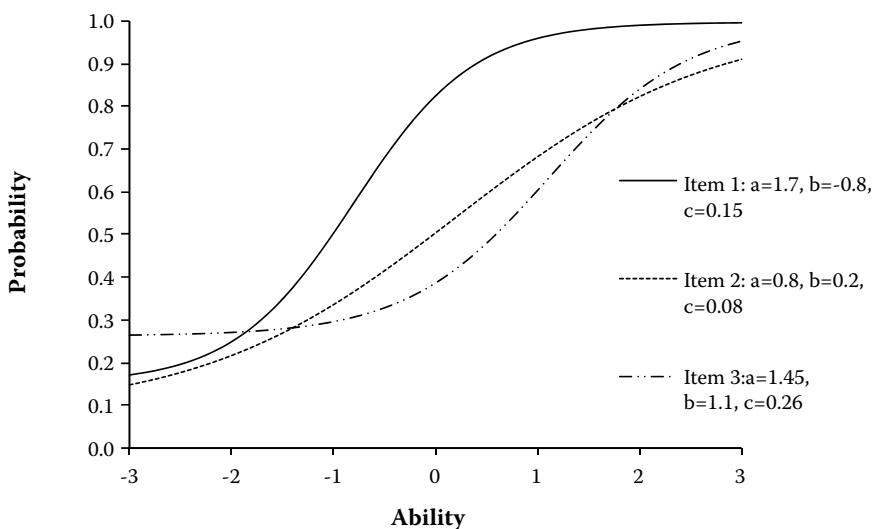


FIGURE 2.2 Illustration of three ICCs from the 3PLM.

the c parameter value and 1.0 (i.e., θ -value associated with $P(\theta) = (1 + c)/2$). The values for the b parameter also typically range from -3.0 to 3.0 , where larger positive values indicate harder items and larger negative values indicate easier items. The discrimination of item i is denoted a_i and is proportional to the slope of the IRF at $\theta = b$ (see Figure 2.2). For “good” items, the a parameter typically ranges from 0.40 to 2.50 . Some testing programs use a version of the model with a scaling constant $D = 1.7$, which was introduced by Birnbaum in 1957 to eliminate scale differences between the item parameters in the two-parameter normal ogive and logistic models. Figure 2.2 graphically illustrates IRFs for three items based on the 3PLM.

The three IRFs shown in Figure 2.2 follow the 3PLM and differ with respect to difficulty, discrimination, and the lower asymptote. The y axis represents the probability of a correct response, $P(\theta)$, while the x axis represents the ability (θ) scale, sometimes called proficiency scale. There are several important features of the 3PLM IRFs shown in Figure 2.2. First, the IRFs are monotonically increasing in that the probabilities increase as the ability levels increase for each item. Second, the IRFs are located throughout the ability distribution, indicating that the items differ in their difficulty. For example, Item 1 is the easiest, Item 2 is moderately difficult, and Item 3 is the hardest ($b_1 = -0.8$, $b_2 = 0.2$, and $b_3 = 1.1$). Third, the inflection point of the respective IRF is located at the b parameter value. Fourth, the a parameter value is proportional to the slope of the IRF at the b parameter value. In addition, the IRF has maximum discriminatory power for examinees whose θ -value is near the b parameter value. For example, Item 1 has maximum discrimination for examinees with θ -values around $b = -0.8$. And fifth, the items differ with respect to the c parameter value as indicated by the disparate lower asymptotes. Interestingly, although Item 3 is generally more difficult

for most examinees throughout the θ scale, lower ability examinees have a higher chance of answering the item correctly compared to Items 1 and 2.

When the c parameter is set equal to zero for an item, Equation 2.1 simplifies to the two-parameter logistic model (2PLM) introduced by Birnbaum (1957, 1958, 1968). This model is more mathematically tractable than the two-parameter normal ogive model introduced by Lord (1952). Since the lower asymptote is fixed at zero, the 2PLM implies that guessing is absent, or at least, negligible for all practical purposes. A dichotomously scored, short-answer item is an example of an item in which the 2PLM is commonly used. Because $c = 0$, the b parameter is now the point on the ability scale at which the probability of a correct response is 0.5.

Constraining the a parameter to be equal to one across all items in Equation 2.1, as well as fixing $c = 0$, produces a third IRT model for dichotomous data known as the one-parameter logistic model (1PLM) or the Rasch model. Neither Lord nor Birnbaum showed any interest in a one-parameter normal ogive or logistic model because of their belief that multiple-choice test items needed at least two parameters in the model to adequately account for actual item response data—one to account for item difficulty and the other to account for item discriminating power. Lord was sufficiently concerned about guessing (and omitted responses) on multiple-choice items (as well as computational demands) that he discontinued his dissertation research (1952, 1953a, 1953b) and pursued research with true score theory instead, until about 1965 (Lord, 1965). He then became convinced that computer power was going to be sufficient to allow him to work with a three-parameter model with the third item parameter in the model to account for the nonzero item performance of low-performing candidates, even on hard test items.

Although the 1PLM produces some attractive statistical features (e.g., exponential family, simple sufficient statistics) when the model actually fits test data, the advantages of the 1PLM come at the expense of assuming the items are all equally discriminating (not to mention free from examinee guessing behavior). Although these assumptions may hold for psychological tests with narrowly defined constructs, they are generally problematic in educational testing. Therefore, the equal discrimination assumption is usually checked closely prior to implementing the 1PLM.

Rasch (1960), on the other hand, developed his one-parameter psychometric model from a totally different perspective than either Lord or Birnbaum. He began with the notion that the odds for an examinee's success on an item depended on the product of two factors—item easiness and examinee ability. Obviously, the easier the item and the more capable the examinee, the higher the odds for a successful response and correspondingly, the higher the probability for the examinee's success on the item. From the definition of odds for success, $P/(1 - P)$, and setting the odds equal to the product of the model parameters for item easiness and examinee ability, it was easy for Rasch to produce a probability model similar to Equation 2.1 with $c = 0.0$ and $a = 1.0$, though in his development, the concepts of item discrimination and guessing

were never considered. At the same time, failure to consider them allows them to become possible sources for model misfit. Rasch certainly would not have worried about guessing in his own work since he developed his model for a longitudinal study with intelligence testing. Also, in 1960, the multiple-choice item was not in use in Denmark.

The three logistic models for analyzing binary data are valuable, and are receiving extensive use in testing practices, but there are several item types or formats in education and psychology that are scored polytomously. For example, many statewide assessments use constructed-response item formats as part of the assessment in which a scoring rubric is implemented to provide partial credit. In addition, Likert type items that provide responses in a graded fashion are commonly used, especially in surveys, questionnaires, and attitudinal inventories. The goal of the IRT models for polytomous data is to describe the probability that an individual responds to a particular category given her or his level of ability and the item properties.

One of the most popular IRT models to address polytomous data, developed by Samejima (1969), is a simple yet elegant extension of the 2PLM and is referred to as Samejima's graded response model (GRM). Samejima was clearly motivated in her work by the fact that all of the modeling up to 1969 was applicable only to binary-scored data. These models were excellent at the time for handling educational testing data where most of the IRT model developers were working (Lord, Novick, Rasch, Wright). However, Samejima was well aware of the use of rating scales (sometimes called ordered response categories by psychologists) and wanted to extend the applicability of item response modeling to these types of data. Samejima's GRM is appropriate for ordered polytomous item responses such as those used in Likert type items or constructed-response items.

For the following explanation, we will consider a five-category (i.e., $K = 5$) item with scores ranging from 0 to 4 (i.e., $k = 0, \dots, 4$). Samejima's work was just the first of many models that followed—including several more of her own (see, for example, Samejima, 1997).

The GRM uses a two-step process in order to obtain the probability that an examinee responds to a particular category. The first step is to model the probability that an examinee's response falls at or above a particular ordered category given θ . The probabilities, denoted $P_{ik}^*(\theta)$, may be expressed as follows:

$$P_{ik}^*(\theta) = \frac{\exp[a_i(\theta_j - b_{ik})]}{1 + \exp[a_i(\theta_j - b_{ik})]} \quad (2.2)$$

In this equation $P_{ik}^*(\theta)$ is referred to as the operating or boundary characteristic function of item i for category k , and indicates the probability of scoring in the k th or higher category on item i (by definition, the probability of responding in or above the lowest category is $P_{ik}^*(\theta) = 1.0$). The a_i parameter refers to the discrimination for item i .

Notational Difference: The formula presented in Equation 2.2 is the same as that presented by Samejima in Chapter 4 and shown below. Samejima’s Equation 4.2 is structured differently, uses different subscripts, and locates the category subscript (x) before (and at a higher level) than the item subscript (g). It is nevertheless functionally and algebraically equivalent to Equation 2.2 in this chapter except for the scaling factor D :

$$P_{x_g}^*(\theta) = \left[1 + \exp \left\{ -D a_g (\theta - b_{x_g}) \right\} \right]^{-1}$$

Note that each item has the same discrimination parameter across all categories in Equation 2.2. Samejima referred to Equation 2.2 as the homogeneous case of her model. She developed a variation on Equation 2.2 in which the operating characteristic function for each score category could vary, and she called this the “heterogeneous” form of the graded response model, but this version of her model did not attract much attention.

Terminology Note: It is common to think of the heterogeneous case as the graded response model with $P_{ik}^*(\theta)$ that vary in shape—as is suggested here. However, the boundary function $P_{ik}^*(\theta)$ actually plays no role in defining heterogeneous models. A fundamental cause of this misunderstanding is the way Samejima uses the term *graded response model*. In this chapter—and in common usage—*graded response model* refers to Samejima’s logistic model in the homogeneous case. However, Samejima uses *graded response model* to refer to a framework that covers all possible polytomous IRT models in both the homogeneous case (including but not limited to the logistic model) and the heterogeneous case, which itself includes many different specific models (see Chapter 4). The most common examples of graded response models in the heterogeneous case are polytomous Rasch models, such as the partial credit model. In such models $P_{ik}^*(\theta)$ is not (and typically cannot be) explicitly modeled and can only be obtained empirically—by summing category probabilities. If this is done, it does indeed transpire that the boundary functions for an item are not parallel. In the terminology used later in this chapter, heterogeneous models are typically direct models.

In Equation 2.2 b_{ik} refers to the ability level in which the probability of responding at or above the particular category equals 0.5 and is often referred to as the threshold parameter. (The threshold parameter is analogous to the item difficulty parameter in the 2PLM.) Since the probability of responding in the first (i.e., lowest) category or higher is defined to be 1.0, the threshold parameter for the first category is not estimated. Therefore, although there are five categories ($K = 5$), there are only four threshold parameters estimated ($K - 1 = 4$). Basically, the item is regarded as a series of $K - 1$ dichotomous responses (e.g., 0 vs. 1, 2, 3, 4; 0, 1 vs. 2, 3, 4; 0, 1, 2 vs. 3, 4; and 0, 1, 2, 3 vs. 4); the 2PLM is used to estimate the IRF for each dichotomy with the added constraint that the slopes are equal within an item.

Once the operating characteristic functions are estimated, the category response functions, which indicate the probability of responding to a particular category given θ , are computed by subtracting adjacent $P_{ik}^*(\theta)$ as follows:

$$P_{ik}(\theta) = P_{ik}^*(\theta) - P_{i(k+1)}^*(\theta) \tag{2.3}$$

By definition, the probability of responding above the highest category is $P_{i(k)}^* = 0.0$; therefore, the probability of responding in the highest category is simply equal to the highest operating characteristic function. For the present example, the category response functions are computed as follows:

$$\begin{aligned}
 P_{i0}(\theta) &= 1.0 - P_{i1}^*(\theta) \\
 P_{i1}(\theta) &= P_{i1}^*(\theta) - P_{i2}^*(\theta) \\
 P_{i2}(\theta) &= P_{i2}^*(\theta) - P_{i3}^*(\theta) \\
 P_{i3}(\theta) &= P_{i3}^*(\theta) - P_{i4}^*(\theta) \\
 P_{i4}(\theta) &= P_{i4}^*(\theta)
 \end{aligned} \tag{2.4}$$

Figures 2.3 and 2.4 illustrate the operating characteristic and category response functions, respectively, for an item with $a = 1.25$, $b_1 = -2.3$, $b_2 = -1.1$, $b_3 = 0.1$, and $b_4 = 1.15$. The figures also highlight some important characteristics of the GRM. First, the operating characteristic curves are ordered from smallest to largest based on the threshold parameters (i.e., $b_1 < b_2 < b_3 < b_4$). Second, the threshold parameters dictate the location of the operating curves. Third, the slope is the same for each of the curves within an item (the slopes are free to vary across items, however). Fourth, for any given value of θ , the sum of the category responses' probabilities equals 1.0. Fifth, the first response category curve is always monotonically decreasing while the last category curve is always monotonically increasing. The middle categories will always be unimodal with the peak located at the midpoint of the adjacent threshold categories.

We will skip ahead a bit to consider a second commonly used IRT model for polytomous data where the data are scored in terms of the number of steps

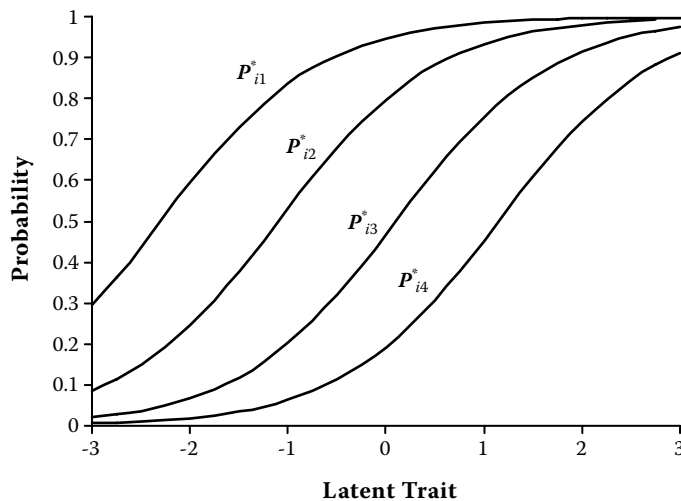


FIGURE 2.3 Operating characteristic curves for a five-category GRM item with $a = 1.25$, $b_1 = -2.3$, $b_2 = -1.1$, $b_3 = 0.1$, and $b_4 = 1.15$.

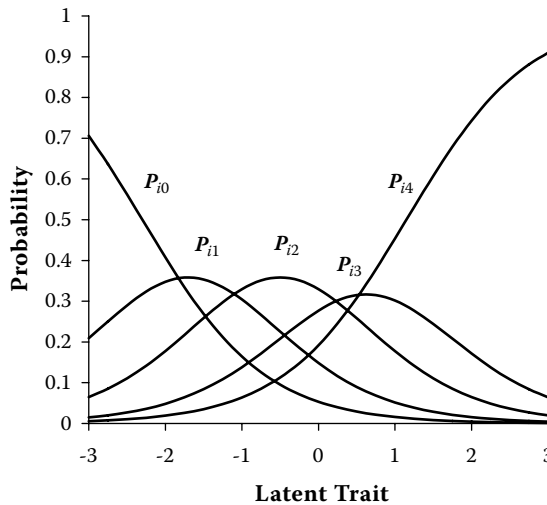


FIGURE 2.4 Category response curves for a five-category GRM item with $a = 1.25$, $b_1 = -2.3$, $b_2 = -1.1$, $b_3 = 0.1$, and $b_4 = 1.15$.

completed in solving a problem (e.g., constructed-response math items): the generalized partial credit model (GPCM; Muraki, 1992, 1993). The model was proposed independently by Yen at the same time, who referred to it as the two-parameter partial credit (2PPC) model and used it in a number of applications to show improved fit relative to the original partial credit model (Fitzpatrick & Yen, 1995; Fitzpatrick et al., 1996).

In contrast to Samejima's GRM, which is considered an indirect model due to the two-step process of obtaining the category response functions, the GPCM is referred to as a direct IRT model because it models the probability of responding to a particular category directly as a function of θ . As the model expression for the GPCM is an exponential divided by a sum of exponentials, it is also classified as a divide-by-total model for the aforementioned reason, while Samejima's GRM is considered a difference model because the category probabilities are based on the difference between the response functions (Thissen & Steinberg, 1986). Muraki was motivated by earlier work from Masters (1982), but Masters did not include a discrimination parameter in his model. This model by Muraki was extensively used at ETS with the National Assessment of Educational Progress (NAEP). With NAEP data, historically, items have varied substantially in their difficulty and discriminating power, and so it was long felt that a two-parameter polytomous response model was needed to fit the data well.

To illustrate the GPCM, we will consider a five-point partial credit item (i.e., $K = 5$) ranging from 0 to 4 (i.e., $k = 0, \dots, 4$). The category response functions, $P_{ik}(\theta)$, for the GPCM may be expressed as follows:

$$P_{ik}(\theta_j) = \frac{\exp\left[\sum_{v=0}^k a_i(\theta_j - b_{iv})\right]}{\sum_{b=0}^{K-1} \exp\left[\sum_{v=0}^b a_i(\theta_j - b_{iv})\right]} \quad (2.5)$$

where $\sum_{v=0}^K (\theta_j - b_{iv}) \equiv 0$.

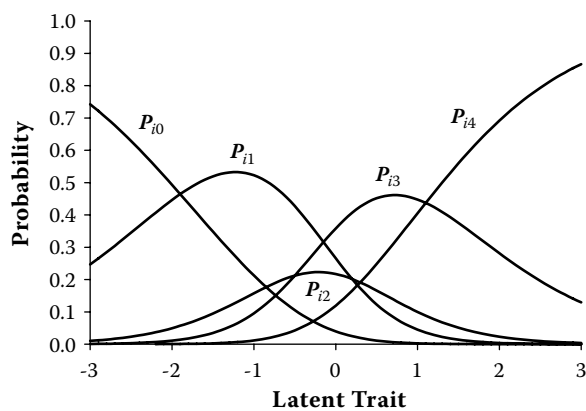


FIGURE 2.5 Category response curves for a five-category GPCM item with $a = 0.99$, $b_1 = -1.9$, $b_2 = 0.2$, $b_3 = -0.5$, and $b_4 = 1.1$.

Notational Difference: The GPCM is also presented in Chapter 3 using the following notation:

$$T(k) = \frac{\exp \sum_{i=0}^k 1.7 a_i (\theta - b_i + d_i)}{\sum_{i=0}^{m-1} \exp \sum_{i=0}^i 1.7 a_i (\theta - b_i + d_i)}$$

Note that in Chapter 3 the overall location parameter (b_i) and the threshold parameters (d_i) are distinct, whereas in Equation 2.5 they are folded together (b_{iv}).

In the GPCM, the b_{iv} are referred to as step difficulty parameters. The step difficulty parameters may be interpreted as representing the difficulty in reaching step k given that the examinee has reached the previous step (i.e., $k - 1$). As a result, b_{iv} s are not necessarily ordered from smallest to largest, in contrast to Samejima’s GRM. As an example, Figure 2.5 illustrates a GPCM item in which the step difficulties are not ordered ($a = 0.99$, $b_1 = -1.9$, $b_2 = 0.2$, $b_3 = -0.5$, and $b_4 = 1.1$).

Terminology Note: In this chapter the authors use the term *step* to describe the boundary parameters in the generalized partial credit model. This term is also used later in this chapter and in Chapter 5, when Masters discusses the partial credit model. Although originally intended to refer to the process of modeling sequential steps toward arriving at a response category, it has been shown that neither model actually operates in this way mathematically (Masters, 1988; Molenaar, 1983). Interpretation of the step difficulty parameters is also complex since each step is modeled in the context of an entire item. A positive consequence of this is that these two models are not restricted to constructed-response item data and can legitimately be used with any polytomous data, including responses to rating scale items. The steps terminology has, however, proved enduring.

As seen from Figure 2.5, the step difficulty parameters represent the value on the θ scale at which two consecutive category response curves intersect (e.g., the curves for category 0 and 1 intersect at $\theta = -1.9$). The relative order of the step difficulty parameters (i.e., intersections) indicates that going from

a 0 to 1 (Step 1) or from 2 to 3 (Step 3) is relatively easy for examinees, while going from 1 to 2 (Step 2) is moderately difficult, and going from 3 to 4 (Step 4) is the most difficult. Furthermore, the effect of the reversal (i.e., Step 2 being more difficult than Step 3) can also be seen in the lower probability of receiving a score of 2 relative to the other categories.

If the a parameter is constrained to be equal across items, the GPCM expressed in Equation 2.5 simplifies to the partial credit model (PCM; Masters, 1982; Masters & Wright, 2004). But the partial credit model was conceived of prior to the generalized partial credit model and by a very different line of reasoning. The partial credit model is an extension of the 1PLM and shares similar attractive statistical properties, such as belonging to an exponential family and simple sufficient statistics. However, the appealing properties of the PCM (and 1PLM/Rasch) can only be realized if the model fits the data. Therefore, the equal item discrimination assumption should be tested empirically before the model is implemented (see, for example, Hambleton & Han, 2005).

Some researchers have been troubled by the fact that in the partial credit model (and the generalized partial credit model) occasional reversals are seen in the step difficulty parameters. Others have argued that the reversals can be interpreted and have chosen not to worry about them (see, for example, Masters, 1982).

Editor Note: In Chapter 4 Andrich discusses the potential pitfalls of “step” reversals and the advantages of being able to model such reversals when they occur in data.

The rating scale model, which has been manifested in a number of ever-increasing flexible forms, can be traced back to the original work of Rasch (1961) and later his doctoral student, Erling Andersen. Then the model became best known in a series of studies by Andrich beginning in 1978 and extended yet again, this time by Muraki in the 1990s with his generalized rating scale model. In the simplest version of this polytomous response model, the thresholds for the score categories differ from each other by an amount that is held constant across items. In addition, there is a shift in these thresholds from one item to the next because the items, in principle, differ in their difficulties. Most of the work associated with the rating scale model has been done under the assumption that all test items are equally discriminating (see Andrich (1988) or Engelhard (2005) for excellent reviews). The exception to this work is the generalized rating scale model introduced by Muraki.

Wright and Masters (1982, chap. 3) provide a very useful comparison of all of the models—rating scale, partial credit, and graded response model—by relating these models to the partial credit model and noting their similarities and differences and deriving one model from another by placing constraints on the models or making additional assumptions. A similar comparison,

which includes the generalized partial credit model, is made in Engelhard (2005). A main difference is that several of the models allow test items to vary in their discriminating powers (GRM and the GPCM include this model parameter).

An interesting comparison between polytomous models is based on Agresti's classification (1990) of response processes (see Mellenbergh, 1995). In adjacent-category models, the examinee is assumed to make his or her response based on a comparison of adjacent categories, and the model represents these comparisons. This type of process underlies the partial credit models. In the graded response model, the basic process is a comparison between the cumulative categories in Equation 2.4, and models of this nature are appropriately called cumulative probability models. The final type of model is the continuation ratio model, which assumes that the examinee basically chooses between a current category and proceeding with one of the higher-order categories. The belief that the last response process is a more adequate description of the examinee's behavior on polytomous items was Verhelst, Glas, and de Vries' (1997) and Tutz's (1997) reason for modifying the PCM.

Nonparametric IRT Models

The previously described models are referred to as parametric because they require the IRF to follow a specific parametric expression. For example, when using the 2PLM, the underlying IRF must follow the logistic expression in Equation 2.2 with parameters a_i and b_i . However, in order to fit the data, for some items on an educational or psychological assessment, the response data may not conform to such an expression and these items would be discarded from a test. For such items, there is a class of nonparametric models based on ordinal assumptions only that may be used to determine the IRF.

While there are several nonparametric methods for modeling the IRF, one of the more popular methods, developed by Ramsay (1991), is kernel regression. For another line of nonparametric modeling of test data, see the work of Molenaar (1997) and Sijtsma and Molenaar (2002) for up-to-date reviews. The essential principle underlying this approach is the replacement of the regression function of the responses on some independent ability score by a function obtained through a smoothing operation. The smoothing operation is based on local weighted averaging of the responses with a so-called kernel as the weight function. More specifically, in order to obtain kernel-smoothed estimates of the response function of item i , the following steps are implemented:

1. Select Q evaluation points along the proposed ability scale, denoted as x_q (e.g., $x_1 = -3.00$, $x_2 = -2.88$, $x_3 = -2.76$, ..., $x_{49} = 2.76$, $x_{50} = 2.88$, $x_{51} = 3.00$, where $Q = 51$).
2. Obtain an ability score that is independent of item i and transform it to have this scale. Usually, the *rest score* is taken (i.e., the total number-correct score that excludes item i). Let X_j denote the score for examinee j .

3. Estimate the IRF using the following formula:

$$\hat{P}_i(x_q) = \sum_{j=1}^N w_{jq} u_{ij} \quad (2.6)$$

where N indicates the number of examinees, u_{ij} is the response by examinee j to item i (i.e., 0 or 1), and w_{jq} represents the weight assigned to examinee j at evaluation point x_q . The weight for examinee j at evaluation point q (i.e., x_q) is calculated as follows:

$$w_{jq} = \frac{K\left(\frac{X_j - x_q}{h}\right)}{\sum_{j=1}^N K\left(\frac{X_j - x_q}{h}\right)} \quad (2.7)$$

where x_q refers to evaluation point q , and X_j is the adopted ability score for examinee j .

Two important components of the formula are bandwidth parameter h and kernel function K . The bandwidth parameter controls the amount of bias and variation in the IRF. As h decreases, the amount of bias is reduced but the variation is increased (i.e., less smoothness). The opposite occurs when h is increased. As a rule of thumb, h is often set equal to $1.1 \cdot N^{-0.2}$ (Ramsay, 1991) so as to produce a smoothed function with acceptable bias. Kernel function K is chosen to be nonnegative and to approach zero as X_j moves away from an evaluation point, x_q . Two commonly used kernel functions are the Gaussian [$K(y) = \exp(-y^2)$] and uniform kernels [$K(y) = 1$ if $|y| \leq 1$, else 0]. Given the previous information, it is apparent that the further an examinee's ability score X_j is away from evaluation point x_q , the less weight that examinee has in determining $\hat{P}(x_q)$. For example, the Gaussian kernel has this feature; its value is largest at $y = 0$, and it decreases monotonically with the distance from this point.

The computer program TESTGRAF (Ramsay, 1992) is often used to perform kernel smoothing on test data. Figure 2.6 illustrates a nonparametrically estimated IRF using TESTGRAF for an item from a large-scale assessment. The psychological meaning for the curve is not clear, but with parametric models, the unusual shape of the curve would not even be known.

It is apparent from Figure 2.6 that the advantage of estimating the IRF nonparametrically is that because it is not constrained to follow a monotonic parametric shape, it provides a deeper understanding of its ordinal features. For example, the 3PLM would never be able to show the local decrease in response probability just above ability equal to -2 in Figure 2.6. A statistical disadvantage of the nonparametric approach, however, is the large number of parameters per item that has to be estimated, and therefore the large amount

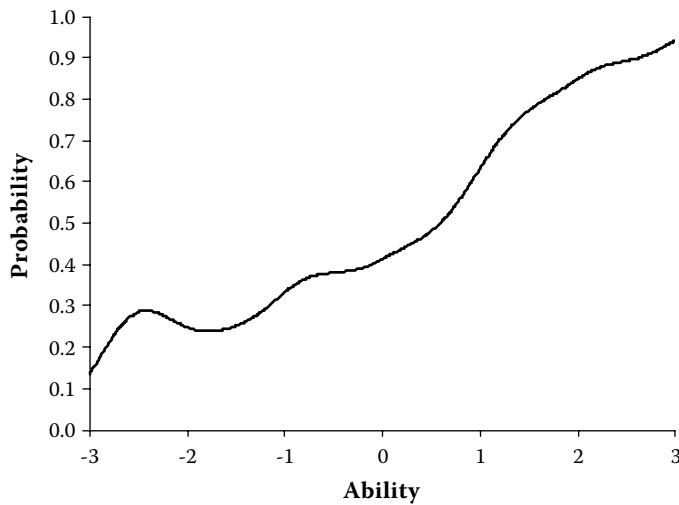


FIGURE 2.6 A nonparametric ICC estimated using TESTGRAF for an item on a large-scale assessment.

of response data required for its application. For example, for the rest score, the number of item parameters in a nonparametric approach is $n - 1$ (namely, one response probability for each possible score minus 1), whereas the 3PLM requires estimation of only three parameters per item. Further, if all items on a test are modeled nonparametrically, then certain applications within IRT cannot be performed. For example, the same ability parameter cannot be estimated from different selection of items, and the nonparametric approach consequently fails to support applications with structurally incomplete designs (e.g., adaptive testing and scale linking). Nevertheless, the nonparametric approach to modeling the IRF has proven to be a useful tool for the psychometric properties of an instrument. For example, nonparametric models have been used for item analysis (Ramsay, 1991; Santor, Zuroff, Ramsay, Cervantes, & Palacios, 1995), testing for differential item functioning (Santor, Ramsay, & Zuroff, 1994; Shealy & Stout, 1993), testing possible local dependencies, and even testing the fit of a parametric model (Douglas & Cohen, 2001; Wells & Bolt, 2008). Therefore, the use of nonparametric models in conjunction with parametric models appears to be a productive strategy for building a meaningful score scale, and we expect to see more applications of nonparametric models to handle polytomous response data in the coming years. The software is available, and the few applications so far appear very promising.

Other IRT Models

Samejima has been prolific over the years in her development of IRT models for handling polytomous response data. Her creativity and insight placed her 15 to 20 years ahead of the time when her models would be needed. Following her work with the graded response model, Samejima (1972) made

the logical jump to the case where continuous response or free response data replaced categorical data, she extended her work to multidimensional modeling of polytomous response data, and in 1997 she extended her modeling work again, this time to incorporate some subtleties in the estimation of ability. Unlike the GRM, the free-response model has received very little attention.

While the present chapter covers the major IRT models for dichotomous and polytomous data, it does not provide a description of all available IRT models. For a thorough description of several other popular IRT models such as other polytomous models (e.g., nominal categories model, rating scale model), multidimensional IRT models for both dichotomous and polytomous data, other nonparametric models (e.g., the monotone homogeneity model), and unfolding models for nonmonotone items, see Fischer (1974), van der Linden and Hambleton (1997), Ostini and Nering (2006), Andrich and Luo (1993), Roberts, Donoghue, and Laughin (2000), and Verhelst and Verstralen (1993).

Parameter Estimation for IRT Models

Perhaps a few words regarding parameter estimation would be useful. Much of the work to date has been for models handling dichotomously scored data—but the maximum likelihood and the Bayesian estimation principles applied equally well to polytomous models, although the complexity is substantially increased because of the increase in the number of model parameters.

The utility of IRT depends on accurately estimating the item parameters and examinee ability. When the ability parameters are known, estimating the item parameters is straightforward. Similarly, estimating the ability parameter is straightforward when the item parameters are known. The challenge, however, is estimating the parameters when both sets are unknown. Due to the complicated nature of the equations for IRT parameter estimation, only a brief description of a few popular methods will follow. See Baker and Kim (2004) for a detailed description of popular estimation methods.

Joint maximum likelihood estimation (JMLE) and marginal maximum likelihood estimation (MMLE; Bock & Lieberman, 1970) are two estimation methods well addressed in the literature. Although MMLE has become the standard of the testing and JMLE lacks a basic statistical requirement, the latter is simpler to describe and is therefore outlined first. JMLE uses an iterative, two-stage procedure to estimate the item and person parameters simultaneously. In Stage 1, the item parameters are estimated assuming the ability parameters are known (simple functions of raw scores may be used as initial estimates of θ) by maximizing the following likelihood function for the responses of N examinees on item i :

$$L(\mathbf{a}, \mathbf{b}, \mathbf{c}; \mathbf{u}, \boldsymbol{\theta}) = \prod_{j=1}^N \prod_{i=1}^n P_j^{u_j} (1 - P_j)^{(1-u_j)} \quad (2.8)$$

where $\mathbf{u}$ represents the matrix with the responses, θ , and $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are the vectors with the examinee and item parameters, respectively, P_j is the model-based probability, and u_j is the item response for examinee j . By taking derivatives and setting them equal to zero, maximum likelihood estimates for each item parameter are obtained, for example, via a Newton-Raphson procedure.

In Stage 2, the ability parameters are estimated treating the item parameters from Stage 1 as known by maximizing the likelihood function for a response pattern for examinee j , which by assuming local independence may be expressed as follows:

$$L(\theta; \mathbf{u}, \mathbf{a}, \mathbf{b}, \mathbf{c}) = \prod_{j=1}^N \prod_{i=1}^n P_j^{u_j} (1 - P_j)^{(1-u_j)} \quad (2.9)$$

where n indicates the number of items. Observe that the likelihood in Equation 2.9 is now treated as a function of the unknown ability parameters given the response data and all item parameters.

Before proceeding back to Stage 1, the updated ability parameter estimates are renormed using the restrictions adopted to eliminate the indeterminacy in the scale (usually, mean estimate equal to zero and standard deviation to one). The two stages are repeated, using the updated estimates from each subsequent stage, until a convergence criterion is met (e.g., estimates change a minimal amount between iterations).

Unfortunately, because the model item and person parameters are being estimated simultaneously in JMLE, the estimates are not consistent. Bock and Lieberman (1970) developed MMLE to address the disadvantages of JMLE by integrating the unknown ability parameters out of the likelihood function so that only the item parameters are left to be estimated. Therefore, the problem becomes one of maximizing a marginal likelihood function in which the unknown ability parameters have been removed through integration. Bock and Aitkin (1981) implemented MMLE using an expectation-maximization (EM) algorithm.

Once the item parameters have been estimated using MMLE, estimates of ability can be derived by treating the item parameter estimates as if they are known item parameters, that is, using the same method outlined in Stage 2 of JMLE. Although for applications with a single standard test without missing data simple number-correct scores may be appropriate, there are several advantages to estimating θ s, including the IRT estimated θ s are comparable when items are added or deleted from the test, they adjust the estimates for the properties of the individual items (such as their difficulty and discrimination), they produce more accurate standard errors, they provide better adjustments for guessing than classical methods, and they are on the same scale as the difficulty parameters. A primary disadvantage of MLE of θ is that for examinees who answer all items correctly or incorrectly, no estimate can be obtained.

An alternative approach for estimating ability is to use the Bayesian approach, in which prior information is provided about the ability parameters in the form of a prior distribution and is incorporated into the likelihood function. The prior distribution is updated by the response data into a posterior distribution for θ , which is the Bayesian estimate of θ . The mean of the posterior distribution may be used as a point estimate of θ known as the expected a posteriori (EAP) estimate. The mode of the posterior distribution may also be used as a point estimate of ability and is known as the maximum a posterior (MAP) estimate (see, e.g., Swaminathan, 2005).

The next generation of IRT modeling for both dichotomous and polytomous response data will likely proceed in a Bayesian way with new numerical procedures for establishing the posterior distribution of model parameters known as Monte Carlo Markov chain (MCMC) procedures. The common feature of the posterior is that they explore the posterior distribution by performing iterative random draws from the posterior distribution of one class of parameters given the previous draws from those of all other parameters. Because of its iterative approach, an MCMC procedure is particularly powerful for models with a high-dimensional parameter space. They also do not require the calculation of the first and second derivatives, which makes MLE cumbersome for complex models. At the same time, MCMC procedures can take several hours for a complex model with a larger data set to produce proper parameter estimates. Still, from a modeling perspective, it allows researchers to be very creative, and currently implementations of MCMC procedures for advanced IRT models are being intensively investigated.

Conclusion

Today, though many polytomous IRT models have been developed, only a handful are receiving frequent use in education and psychology: (1) the nominal response model, (2) the graded response model, (3) the polytomous Rasch model, (4) the partial credit model, and (5) the generalized partial credit model. Of the unidimensional models for handling polytomous response data, these are almost certainly the five most frequently used, and the ones that will be described in greater detail in subsequent chapters.

One of the biggest challenges facing applications of polytomous item response models has been the shortage of user-friendly software. Several software packages are available for parameter estimation: BILOG-MG (www.ssicentral.com) can be used with the one-, two-, and three-parameter logistic models; PARSCALE (www.ssicentral.com) with the graded response model and the generalized partial credit model; MULTILOG (www.ssicentral.com) with the nominal response model and the graded response model, and WINSTEPS and FACETS (www.winsteps.com) and CONQUEST with the dichotomous Rasch model, partial credit model, and polytomous Rasch model. The Web site www.assess.com is particularly

helpful is locating IRT software. We are already aware, too, of several major releases of software scheduled for 2010, including a new version of Multilog and software to comprehensively address model fit. Still, in general, it is to be hoped that software in the future can be made more user-friendly.

Approaches to model fit too remain as technical challenges. Testing a model with data is a losing proposition because with a sufficiently large sample size (which is desirable for obtaining stable model parameter estimates), power to reject any IRT model is high. Investigating practical consequences of any model misfit remains a more promising direction for studies of model fit. Alternatively, it is becoming more standard to test one model against another. This topic is generating considerable interest currently.

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